

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Scenario-Based

Code of Conduct:

192472056
 I V. Sheersbika (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
 I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Scenario

A smart Campus has been allocated the IP block 192.168.100.0/24. Different departments require different subnet sizes.

Department	Subnet Mask
School of Computing	/25
School of Electronics	/26
School of Mechanical Engineering	/27

Subnet Allocation.

1) School of Computing (120)

Subnet Mask

255.255.255.128

Block size

128

Network Address

192.168.100.0

Broadcast Address

192.168.100.127

Valid Host Range

192.168.100.1 - 192.168.100.126

Usable hosts : $2^{(32-25)} - 2 = 2^7 - 2 = 126$

2) School of Electronics

Subnet Mask : 255.255.255.192

Block size : 64

Network address : 192.168.100.192

Broadcast Address : 192.168.100.191

Host Range : 192.168.100.129 - 192.168.100.190

Usable Hosts : $2^6 - 2 = 62$

3) School of Mechanical Engineering (127)

Subnet Mask : 255.255.255.224

Block size : 32

Network Address : 192.168.100.192

Broadcast Address : 192.168.100.223

Host Range : 192.168.100.193 - 192.168.100.222

Usable Hosts : $2^5 - 2 = 30$

4) Remaining Unused IP Range

Original Block : 192.168.100.0 - 192.168.100.255
Allocated,

→ 192.168.100.0 - 127

→ 192.168.100.128 - 191

→ 192.168.100.192 - 223

Remaining : 192.168.100.224 - 192.168.100.255

Remaining capacity : 32 Addresses

2) Scenario

Company Network , 10.10.0.0/16

1) Software Development (124)

Subnet Mask : 255.255.255.0

Network Address : 10.10.0.0

Broadcast Address : 10.10.0.255

Host Range : 10.10.0.1 - 10.10.0.254

Usable Hosts : $2^8 - 2 = 254$

2) Quality Assurance (130)

Network Address : 10.10.1.0

Broadcast : 10.10.1.127

Host Range : 10.10.1.1 - 10.10.1.126

Usable Hosts : 126

3) Human Resources (126)

Network Address : 10.10.1.128

Broadcast Address : 10.10.1.191

Host Range : 10.10.1.129 - 10.10.1.190

Usable Hosts : 62

4) Executive Management (127)

Network Address : 10.10.1.192

Broadcast Address : 10.10.1.223

Host Range : 10.10.1.193 - 10.10.1.222

Usable Hosts : 30

Unused Address Range

Allocated ; 10.10.00 - 10.10.1.223 .

Unused : 10.10.1.224 - 10.10.255.255

3) IPv6 address

2001:0db8:abcd:0000:0000:0000:0000:0000 / 64

1) Compressed form

Leading zeros removed

consecutive zero blocks replaced by ::

→ 2001:db8:abcd::/64

2) Expanded form

2001:0db8:abcd:0000:0000:0000:0000:0000

3) Network prefix

Prefix length : /64

Network prefix : 2001:db8:abcd:0000::/64

Host portion : Last 64 bits.

4) Number of Host Addresses

Remaining Bits : $128 - 64 = 64$

Hosts = 2^{64}

= 18446744 073709551 616

5) Total Possible Hosts

IPv6 has no broadcast address

Therefore,

Total Host Address. = 2^{64} .

4) Routing Table

Destination Network

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

Destination IP

192.168.2.45

1) Destination Subnet

192.168.2.45 belongs to 192.168.2.0/24

2) Matching Routing Entry

192.168.2.0/24

3) Next Hop

Packet is forwarded through the interface connected to 192.168.2.0/24 Network.

4) Default Route

If no entry matches Default Route: 0.0.0.0/0
The router forwards the packet to the configured default gateway.

5) Administration LAN

192.168.10.0/26

Finance LAN : 192.168.10.64/26

Gateways Administration : 192.168.10.1

Finance : 192.168.10.65

Administration LAN

Network : 192.168.10.0

Broadcast : 192.168.10.63

Host Range : 192.168.10.1 - 192.168.10.62

Suggested PCs

PC1 → 192.168.10.82

PC2 → 192.168.10.3

PC3 → 192.168.10.4

Gateway : 192.168.10.1

Finance LAN

Network : 192.168.10.64

Broadcast : 192.168.10.127

Host Range : 192.168.10.65 - 192.168.10.126

Suggested PCs.

PC1 → 192.168.10.66

PC2 → 192.168.10.67

PC3 → 192.168.10.68

② Gateway : 192.168.10.65